POWERBI WEEK 8 REPORT

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DEPT: BSC DS AND AI 2ND YEAR

AIM

TO ANALYZE STUDENT PERFORMANCE USING MICROSOFT POWER BI BY CREATING VISUALIZATIONS BASED ON STUDENT SCORES, GENDER, RACE/ETHNICITY, PARENTAL EDUCATION, LUNCH TYPE, AND TEST PREPARATION COURSE.

OBJECTIVES

- TO ANALYZE STUDENT PERFORMANCE USING DATA VISUALIZATION.
- TO COMPARE STUDENT SCORES ACROSS DIFFERENT CATEGORIES.
- TO ANALYZE OVERALL PASS PERCENTAGES.
- TO EXAMINE THE RELATIONSHIP BETWEEN TEST PREPARATION AND ACADEMIC PERFORMANCE.
- TO CREATE AN INTERACTIVE DASHBOARD USING MICROSOFT POWER BI.

DATASET DESCRIPTION

THE STUDENT PERFORMANCE DATASET CONTAINS INFORMATION ABOUT STUDENTS, INCLUDING GENDER, RACE/ETHNICITY, PARENTAL LEVEL OF EDUCATION, LUNCH TYPE, TEST PREPARATION COURSE, AND SCORES IN MATHEMATICS, READING, AND WRITING.

THE DATASET WAS USED TO ANALYZE ACADEMIC PERFORMANCE AND IDENTIFY PATTERNS ACROSS DIFFERENT STUDENT CATEGORIES.

DASHBOARD OVERVIEW

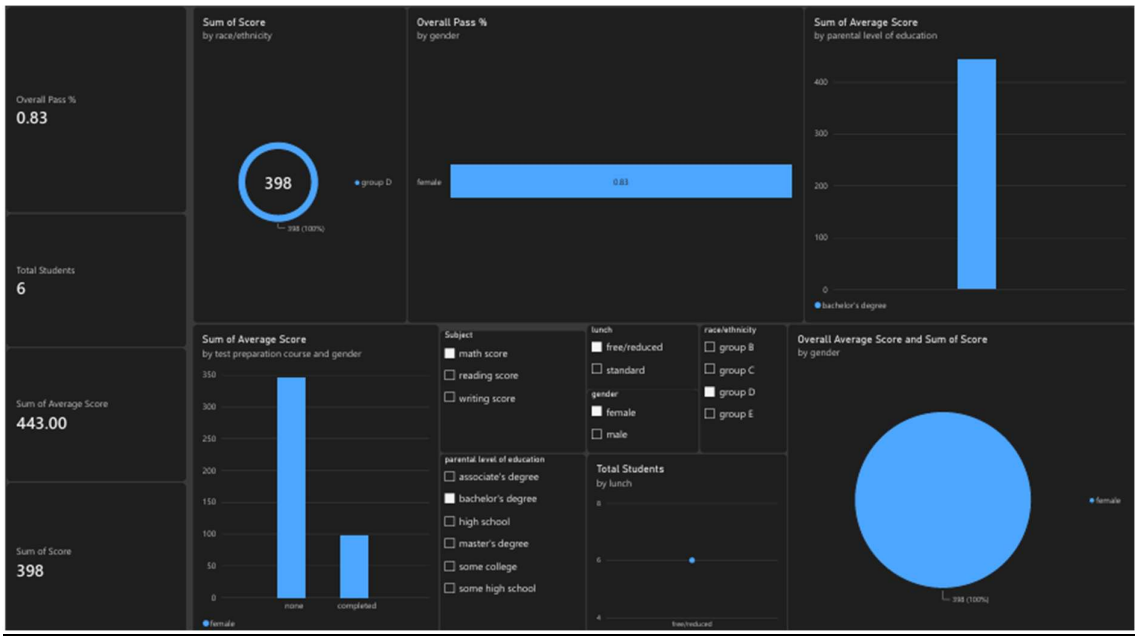
THE STUDENT PERFORMANCE ANALYSIS DASHBOARD WAS CREATED USING MICROSOFT POWER BI. IT COMBINES MULTIPLE

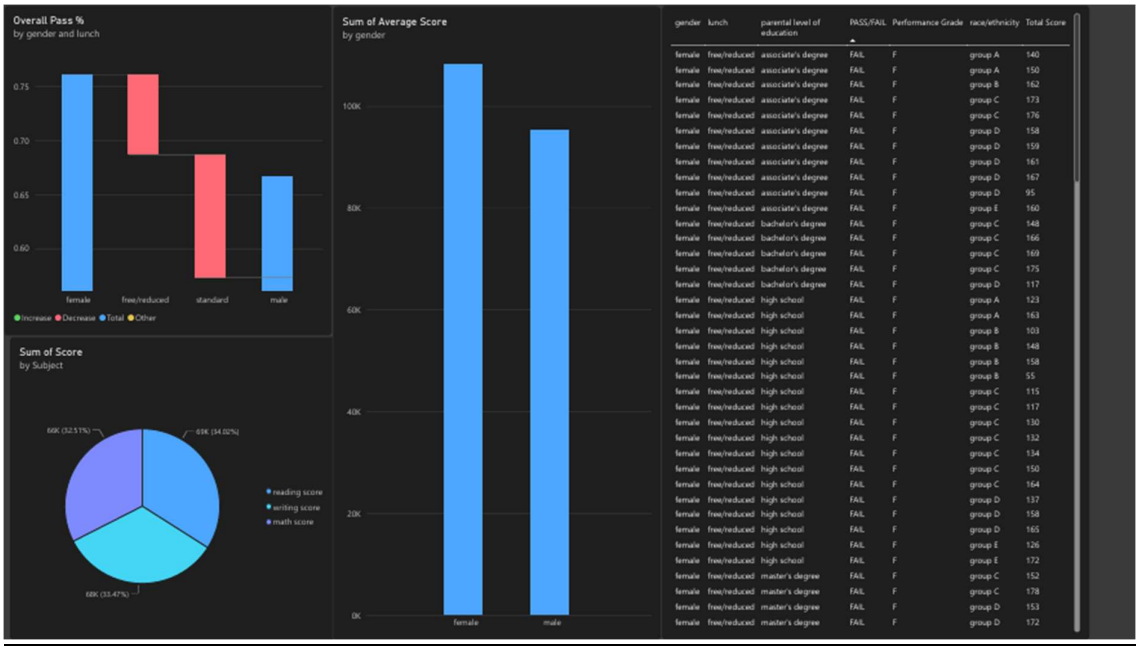
VISUALIZATIONS, SUMMARY CARDS, FILTERS, AND A DETAILED TABLE TO PRESENT STUDENT PERFORMANCE INFORMATION IN AN INTERACTIVE FORMAT.

THE DASHBOARD INCLUDES:

- STUDENT SCORE ANALYSIS BY RACE/ETHNICITY.
- OVERALL PASS PERCENTAGE BY GENDER.
- STUDENT SCORES BY PARENTAL LEVEL OF EDUCATION.
- SCORE COMPARISON BASED ON TEST PREPARATION COURSE AND GENDER.
- PASS PERCENTAGE ANALYSIS BY GENDER AND LUNCH TYPE.
- SUBJECT-WISE SCORE ANALYSIS.
- INTERACTIVE SLICERS FOR FILTERING THE DATA.
- A DETAILED STUDENT PERFORMANCE TABLE.

VISUALIZATIONS USED AND WORK PERFORMED





1 BAR CHART – SUM OF SCORE BY RACE/ETHNICITY

VISUALIZATION USED: BAR CHART

WORK PERFORMED:
A BAR CHART WAS USED TO REPRESENT THE SUM OF STUDENT SCORES ACROSS DIFFERENT RACE/ETHNICITY GROUPS. THIS VISUALIZATION HELPS COMPARE THE TOTAL SCORES AMONG THE AVAILABLE CATEGORIES.

2 CARD – OVERALL PASS PERCENTAGE

VISUALIZATION USED: CARD

WORK PERFORMED:
A CARD VISUAL WAS USED TO DISPLAY THE OVERALL PASS PERCENTAGE. IT PROVIDES A QUICK SUMMARY OF STUDENT PASSING PERFORMANCE IN THE DASHBOARD.

3 CHART – SUM OF AVERAGE SCORE BY PARENTAL LEVEL OF EDUCATION

VISUALIZATION USED: CHART

WORK PERFORMED:
A CHART WAS CREATED TO DISPLAY STUDENT SCORE VALUES BASED ON PARENTAL LEVEL OF EDUCATION. THIS HELPS COMPARE THE DISPLAYED SCORES ACROSS DIFFERENT EDUCATIONAL BACKGROUNDS.

4 CHART – AVERAGE SCORE BY TEST PREPARATION COURSE AND GENDER

VISUALIZATION USED: BAR CHART

WORK PERFORMED:
A CHART WAS CREATED TO COMPARE STUDENT SCORES BASED ON TEST PREPARATION COURSE COMPLETION AND GENDER. THE CATEGORIES INCLUDE STUDENTS WHO COMPLETED THE COURSE AND THOSE WHO DID NOT COMPLETE IT.

5 CHART – OVERALL PASS PERCENTAGE BY GENDER AND LUNCH TYPE

VISUALIZATION USED: CHART

WORK PERFORMED:
A VISUALIZATION WAS CREATED TO COMPARE OVERALL PASS PERCENTAGES ACROSS GENDER AND LUNCH CATEGORIES. THIS HELPS EXAMINE DIFFERENCES BETWEEN THE DISPLAYED GROUPS.

6 CHART – SUBJECT-WISE SCORE ANALYSIS

VISUALIZATION USED: CHART

WORK PERFORMED:
A SUBJECT-WISE VISUALIZATION WAS CREATED TO COMPARE THE TOTAL SCORES IN MATHEMATICS, READING, AND WRITING. THIS PROVIDES AN OVERVIEW OF THE SCORE DISTRIBUTION ACROSS THE THREE SUBJECTS.

7 SLICERS – INTERACTIVE DATA FILTERING

VISUALIZATION USED: SLICERS

WORK PERFORMED:
INTERACTIVE SLICERS WERE ADDED FOR LUNCH TYPE,

RACE/ETHNICITY, GENDER, SUBJECT, AND PARENTAL LEVEL OF EDUCATION. THESE SLICERS ALLOW USERS TO FILTER THE DASHBOARD AND ANALYZE SELECTED CATEGORIES.

8 TABLE – STUDENT PERFORMANCE DETAILS

VISUALIZATION USED: TABLE

WORK PERFORMED:
A TABLE WAS CREATED TO DISPLAY STUDENT INFORMATION, INCLUDING GENDER, LUNCH TYPE, PARENTAL LEVEL OF EDUCATION, PASS/FAIL STATUS, PERFORMANCE GRADE, RACE/ETHNICITY, AND TOTAL SCORE. THIS PROVIDES A DETAILED VIEW OF THE STUDENT RECORDS.

SUMMARY OF WORK PERFORMED

THE STUDENT PERFORMANCE ANALYSIS DASHBOARD WAS DEVELOPED USING MICROSOFT POWER BI. THE WORK INVOLVED IMPORTING THE DATASET, ORGANIZING THE AVAILABLE STUDENT ATTRIBUTES, CREATING CHARTS AND SUMMARY CARDS, ADDING INTERACTIVE SLICERS, AND PRESENTING STUDENT INFORMATION IN A DETAILED TABLE.

THE DASHBOARD COMBINES MULTIPLE VISUALIZATIONS TO HELP USERS EXPLORE STUDENT SCORES, COMPARE PERFORMANCE ACROSS DIFFERENT CATEGORIES, AND EXAMINE ACADEMIC PERFORMANCE INTERACTIVELY.

CONCLUSION

THIS PROJECT PROVIDED PRACTICAL EXPERIENCE IN IMPORTING DATA, PREPARING DATASETS, CREATING VISUALIZATIONS, AND DEVELOPING AN INTERACTIVE DASHBOARD USING MICROSOFT POWER BI.

THE DASHBOARD HELPS USERS EXPLORE STUDENT PERFORMANCE ACROSS DIFFERENT DEMOGRAPHIC AND

ACADEMIC CATEGORIES. IT DEMONSTRATES HOW DATA VISUALIZATION CAN BE USED TO ORGANIZE INFORMATION, COMPARE PERFORMANCE, AND UNDERSTAND PATTERNS IN A DATASET.